



Manajemen Proyek Sistem Informasi

Project Management Process and Integration

S1 SISTEM INFORMASI
FAKULTAS REKAYASA INDUSTRI
TELKOM UNIVERSITY





Learning Objectives

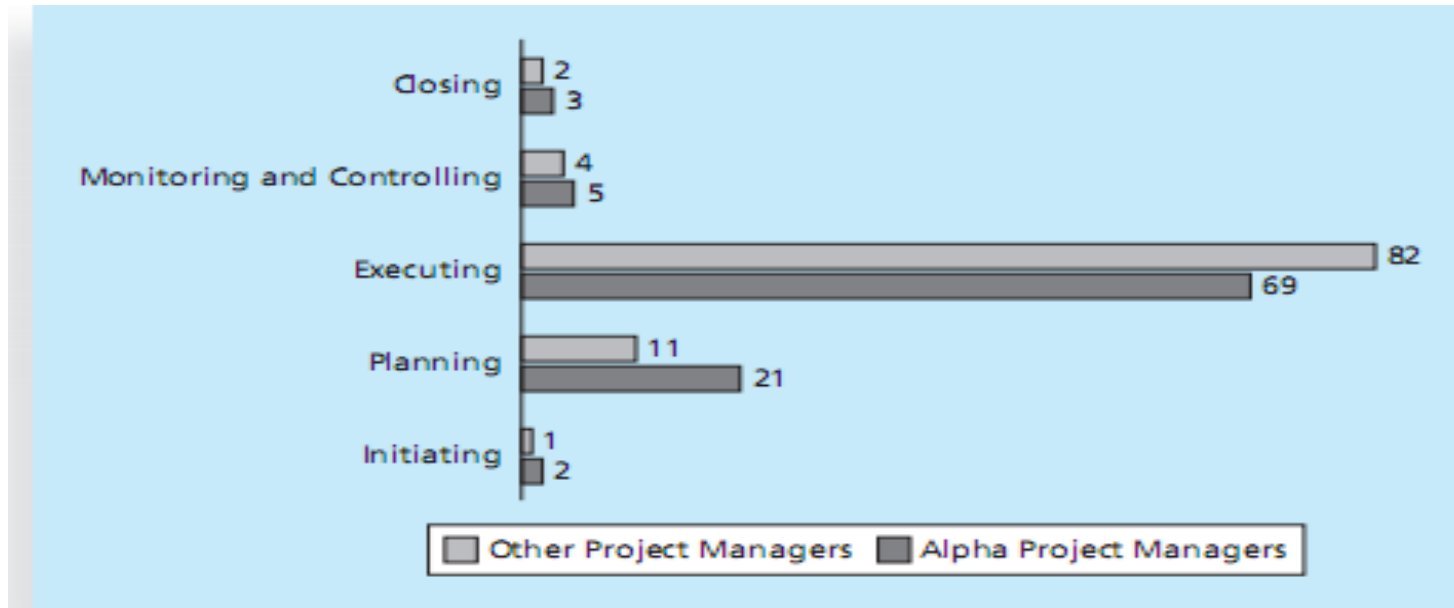
- Describe the **five project management (PM) process groups**, the typical level of activity for each, and the interactions among them
- Understand **how the PM process groups relate to the PM knowledge areas**
- Discuss **how organizations develop information technology PM methodologies to meet their needs**
- Review a **case study of an organization applying the project management process groups to manage an information technology project**; describe **outputs of each process group**; and understand the contribution that effective project initiating, planning, executing, monitoring and controlling, and closing make to project success
- Review the **same case study of a project managed with an agile focus** to illustrate the key differences in approaches
- Describe **several templates for creating documents** for each **process group**



Project Management Process Groups

- A **process** is a **series of actions** directed toward a particular result
- **Project management** can be viewed as a **number of interlinked processes**
- The **project management process groups** include:
 1. **Initiating** processes
 2. **Planning** processes
 3. **Executing** processes
 4. **Monitoring and controlling** processes
 5. **Closing** processes

Percentage of Time Spent on Each Process Group



Source: Andy Crowe



What Went Wrong?

- Philip A. Pell, PMP, commented on **how the U.S. Internal Revenue Service (IRS) needed to improve its project management process**. “Pure and simple, good, methodology-centric, predictable, and repeatable project management is the SINGLE greatest factor in the success (or in this case failure) of any project... The project manager is ultimately responsible for the success or failure of the project.”*
- In 2006, the **IRS lost more than \$320 million due to a botched fraud-detection system** project
- A 2008 U.S. Government Accountability Office (GAO) report stated that **IRS had fixed just 29 of 98 information security weaknesses** identified the previous year.



Media Snapshot

- Just as information technology projects need to **follow the project management process groups**, so do other projects, such as the **production of a movie**
- Processes involved in making movies might include screenwriting (**initiating**), producing (**planning**), acting and directing (**executing**), editing (**monitoring and controlling**), and releasing the movie to theaters (**closing**)
- Many people enjoy watching the extra features on a DVD that describe how these processes lead to the creation of a movie... This acted “...not as promotional filler but as a serious and meticulously detailed examination of the entire filmmaking process.”*
- Project managers in any field know **how important it is to follow a good process**

Jacks, Brian, “Lord of the Rings: The Two Towers Extended Edition (New Line)”, Underground Online (accessed from www.ugo.com August 4, 2004)



Mapping the Process Groups to the Knowledge Areas

- You can map the main activities of each PM process group into the knowledge areas using the PMBOK® Guide 6th ED
- Note that there are activities from each knowledge area under the planning and monitoring and controlling process groups
- There are 47 Processes in 10 Knowledges

Project Management Process Groups and Knowledge Area Mapping (1)

Knowledge Area	Project Management Process Groups				
	Initiating	Planning	Executing	Monitoring and Controlling	Closing
<i>Project Integration Management</i>	Develop project charter	Develop project management plan	Direct and manage project work	Monitor and control project work, Perform integrated change control	Close project or phase
<i>Project Scope Management</i>		Plan scope management, Collect requirements, Define scope, Create WBS		Validate scope, Control scope	
<i>Project Time Management</i>		Plan schedule management, Define activities, Sequence activities, Estimate activities resources, Estimate activity durations, Develop schedule		Control schedule	

Project Management Process Groups and Knowledge Area Mapping (2)

Knowledge Area	Project Management Process Groups				
	Initiating	Planning	Executing	Monitoring and Controlling	Closing
<i>Project Cost Management</i>		Plan cost management, Estimate costs, Determine budget		Control costs	
<i>Project Quality Management</i>		Plan quality management	Perform quality assurance	Control quality	
<i>Project Human Resource Management</i>		Plan human resource management	Acquire project team, Develop project team, Manage project team		
<i>Project Communications Management</i>		Plan communications management	Manage communications	Control communications	
<i>Project Risk Management</i>		Plan risk management, Identify risks, Perform qualitative risk analysis, Perform quantitative risk analysis, Plan risk responses		Control risks	
<i>Project Procurement Management</i>		Plan procurement management	Conduct procurements	Control procurements	Close procurements
<i>Project Stakeholder Management</i>	Identify stakeholders	Plan stakeholder management	Manage stakeholder engagement	Control stakeholder engagement	

Organizational Structure Influences on Projects

Project Characteristics	Organizational Structure Type				
	Functional	Weak Matrix	Balanced Matrix	Strong Matrix	Project
Project manager's authority	Little or none	Limited	Low to moderate	Moderate to high	High to almost total
Percent of organization's personnel assigned full-time to project work	Virtually none	0–25%	15–60%	50–95%	85–100%
Who controls the project budget	Functional manager	Functional manager	Mixed	Project manager	Project manager
Project manager's role	Part-time	Part-time	Full-time	Full-time	Full-time
Common title for project manager's role	Project coordinator/ project leader	Project coordinator/ project leader	Project manager/ project officer	Project manager/ program manager	Project manager/ program manager
Project management administrative staff	Part-time	Part-time	Part-time	Full-time	Full-time



Developing an IT Project Management Methodology

- Many organizations develop their **own project management methodologies**, especially for IT projects
- A **methodology** describes **how things should be done**; a standard describes what should be done
- **PRINCE2, Agile, RUP**, and **Six Sigma** provide different **project management methodologies**



What Went Right?

- **AgênciaClick**, an interactive advertising and online communications company based in São Paulo, Brazil, made **PMI's list of outstanding organizations in project management in 2007**
- Since 2002, the company saw **revenues jump 132 percent**, primarily **due to their five-year emphasis on practicing good project management** across the entire company



Case Study: JWD Consulting's Project Management Intranet Site

- This case study provides an example of **what's involved in initiating, planning, executing, controlling, and closing** an IT project
- Note: This case study provides a **big picture view of managing a project**; **later chapters provide detailed information on each knowledge area**



Project Pre-initiation

- It is good practice to **lay the groundwork for a project before it officially starts**
- Senior managers often perform several **pre-initiation tasks**, including the following:
 1. Determine the **scope, time, and cost** constraints for the project
 2. Identify the **project sponsor**
 3. Select the **project manager**
 4. Develop a **business case** for a project (see Table 3-2 for an example)
 5. Meet with the project manager to **review the process and expectations** for managing the project
 6. Determine if the **project should be divided into two or more** smaller projects



Project Pre-initiation: Business Case

1. Introduction/Background
2. Business Objective
3. Current Situation and Problem/Opportunity Statement
4. Critical Assumption and Constraints
5. Analysis of Option and Recommendation
6. Preliminary Project Requirements
7. Budget Estimate and Financial Analysis
8. Schedule Estimate
9. Potential Risks
10. Exhibits



Project Pre-initiation: Business Case (2)

1.0 Introduction/Background

JWD Consulting's core business goal is to provide world-class project management consulting services to various organizations. The CEO, Joe Fleming, believes the firm can streamline operations and increase business by providing information related to project management on its intranet site, making some information and services accessible to current and potential clients.

2.0 Business Objective

JWD Consulting's strategic goals include continuing growth and profitability. The project management intranet site project will support these goals by increasing visibility of the firm's expertise to current and potential clients by allowing client and public access to some sections of the intranet. The project will also improve profitability by reducing internal costs by providing standard tools, techniques, templates, and project management knowledge to all internal consultants. Because JWD Consulting focuses on identifying profitable projects and measuring their value after completion, this project must meet those criteria.

3.0 Current Situation and Problem/Opportunity Statement

JWD Consulting has a corporate Web site as well as an intranet. The firm currently uses the Web site for marketing information. The primary use of the intranet is for human resource information, such as where consultants enter their hours on various projects, change and view their benefits information, and access an online directory and Web-based e-mail system. The firm also uses an enterprise-wide project management system to track all project information, focusing on the status of deliverables and meeting scope, time, and cost goals. There is an opportunity to provide a new section on the intranet dedicated to sharing consultants' project management knowledge across the organization. JWD Consulting only hires experienced consultants and gives them freedom to manage projects as they see fit. However, as the business grows and projects become more complex, even experienced project managers are looking for suggestions on how to work more effectively.



Project Pre-initiation: Business Case (3)

4.0 Critical Assumptions and Constraints

The proposed intranet site must be a valuable asset for JWD Consulting. Current consultants and clients must actively support the project, and it must pay for itself within one year by reducing internal operating costs and generating new business. The Project Management Office manager must lead the effort, and the project team must include participants from several parts of the company, as well as from current client organizations. The new system must run on existing hardware and software, and it should require minimal technical support. It must be easily accessible by consultants and clients and be secure from unauthorized users.

5.0 Analysis of Options and Recommendation

There are three options for addressing this opportunity:

1. Do nothing. The business is doing well, and we can continue to operate without this new project.
2. Purchase access to specialized software to support this new capability with little in-house development.
3. Design and implement the new intranet capabilities in-house, using mostly existing hardware and software.

Based on discussions with stakeholders, we believe that option 3 is the best option.

6.0 Preliminary Project Requirements

The main features of the project management intranet site include the following:

1. Access to several project management templates and tools. Users must be able to search for templates and tools, read instructions for using these templates and tools, and see examples of how to



Project Pre-initiation: Business Case (4)

apply them to real projects. Users must also be able to submit new templates and tools, which should first be screened or edited by the Project Management Office.

2. Access to relevant project management articles. Many consultants and clients sense an information overload when they research project management information. They often waste time they should be spending with their clients. The new intranet should include access to several important articles on various project management topics, which are searchable by topic, and allow users to ask the Project Management Office staff to find additional articles to meet their needs.
3. Links to other, up-to-date Web sites, with brief descriptions of the main features of the external sites.
4. An “Ask the Expert” feature to help build relationships with current and future clients and share knowledge with internal consultants.
5. Appropriate security to make the entire intranet site accessible to internal consultants and certain sections accessible to others.
6. The ability to charge money for access to some information. Some of the information and features of the intranet site should prompt external users to pay for the information or service. Payment options should include a credit card option or similar online payment transactions. After the system verifies payment, the user should be able to access or download the desired information.
7. Other features suggested by users, if they add value to the business.

Project Pre-initiation: Business Case (5)

7.0 Budget Estimate and Financial Analysis

A preliminary estimate of costs for the entire project is \$140,000. This estimate is based on the project manager working about 20 hours per week for six months and other internal staff working a total of about 60 hours per week for six months. The customer representatives would not be paid for their assistance. A staff project manager would earn \$50 per hour. The hourly rate for the other project team members would be \$70 per hour, because some hours normally billed to clients may be needed for this project. The initial cost estimate also includes \$10,000 for purchasing software and services from suppliers. After the project is completed, maintenance costs of \$40,000 are included for each year, primarily to update the information and coordinate the “Ask the Expert” feature and online articles.

Projected benefits are based on a reduction in hours that consultants spend researching project management information, appropriate tools, and templates. Projected benefits are also based on a small increase in profits due to new business generated by this project. If each of 400 consultants saved just 40 hours each year (less than one hour per week) and could bill that time to other projects that generate a conservative estimate of \$10 per hour in *profits*, then the projected benefit would be \$160,000 per year. If the new intranet increased business by just 1 percent, using past profit information, increased profits due to new business would be at least \$40,000 each year. Total projected benefits, therefore, are about \$200,000 per year. Exhibit A summarizes the projected costs and benefits and shows the estimated net present value (NPV), return on investment (ROI), and year in which payback occurs. It also lists assumptions made in performing this preliminary financial analysis. All of the financial estimates are very encouraging. The estimated payback is within one year, as requested by the sponsor. The NPV is \$272,800, and the discounted ROI based on a three-year system life is excellent at 112 percent.

8.0 Schedule Estimate

The sponsor would like to see the project completed within six months, but there is some flexibility in the schedule. We also assume that the new system will have a useful life of at least three years.

9.0 Potential Risks

This project carries several risks. The foremost risk is a lack of interest in the new system by our internal consultants and external clients. User inputs are crucial for populating information into this system and realizing the potential benefits from using the system. There are some technical risks in choosing the type of software used to search the system, check security, process payments, and so on, but the features of this system all use proven technologies. The main business risk is investing the time and money into this project and not realizing the projected benefits.

Project Pre-initiation: Business Case (6)

10.0 Exhibits

Exhibit A: Financial Analysis for Project Management Intranet Site Project

Discount rate	8%				
Assume the project is done in about 6 months					
	Year				
	0	1	2	3	Total
Costs	140,000	40,000	40,000	40,000	
Discount factor	1	0.93	0.86	0.79	
Discounted costs	140,000	37,037	34,294	31,753	243,084
Benefits	0	200,000	200,000	200,000	
Discount factor	1	0.93	0.86	0.79	
Discounted benefits	0	186,185	171,468	158,766	515,419
Discounted benefits - costs	(140,000)	148,148	137,174	127,013	
Cumulative benefits - costs	(140,000)	8,148	145,322	272,336	← NPV
Payback in Year 1					
Discounted life cycle ROI----->	112%				
Assumptions					
Costs	# hours				
PM (500 hours, \$50/hour)	25,000				
Staff (1500 hours, \$70/hour)	105,000				
Outsourced software and services	10,000				
Total project costs (all applied in year 0)	140,000				
Benefits					
# consultants	400				
Hours saved	40				
\$/hour profit	10				
Benefits from saving time	160,000				
Benefits from 1% increase in profits	40,000				
Total annual projected benefits	200,000				



Project Initiation

- Initiating a project includes recognizing and starting a new project or project phase
- The main goal is to formally select and start off projects

Knowledge Area	Initiating Process	Outputs
<i>Project Integration Management</i>	Develop project charter	Project charter
<i>Project Stakeholder Management</i>	Identify stakeholders	Stakeholder register



Stakeholder Register

Name	Position	Internal/ External	Project Role	Contact Information
Joe Fleming	CEO	Internal	Sponsor	joe_fleming@jwdconsulting.com
Erica Bell	PMO Director	Internal	Project manager	erica_bell@jwdconsulting.com
Michael Chen	Team member	Internal	Team member	michael_chen@jwdconsulting.com
Kim Phuong	Business analyst	External	Advisor	kim_phuong@client1.com
Louise Mills	PR Director	Internal	Advisor	louise_mills@jwdconsulting.com



Stakeholder Management Strategy

Name	Level of Interest	Level of Influence	Potential Management Strategies
Joe Fleming	High	High	Joe likes to stay on top of key projects and make money. Have a lot of short, face-to-face meetings and focus on achieving the financial benefits of the project.
Louise Mills	Low	High	Louise has a lot of things on her plate, and she does not seem excited about this project. She may be looking at other job opportunities. Show her how this project will help the company and her resume.



Project Charters and Kick-off Meetings

- Charters are normally short and include key project information and stakeholder signatures
- It's good practice to hold a kick-off meeting at the beginning of a project so that stakeholders can meet each other, review the goals of the project, and discuss future plans.



Kick Off Meeting

Kick-Off Meeting

[Date of Meeting]

Project Name: Project Management Intranet Site Project

Meeting Objective: Get the project off to an effective start by introducing key stakeholders, reviewing project goals, and discussing future plans

Agenda:

- Introductions of attendees
- Review of the project background
- Review of project-related documents (i.e., business case, project charter)
- Discussion of project organizational structure
- Discussion of project scope, time, and cost goals
- Discussion of other important topics
- List of action items from meeting

Action Item	Assigned To	Due Date

Date and time of next meeting:





Project Planning

- The main purpose of project planning is **to guide execution**
- **Every knowledge area includes planning** information
- **Key outputs** include:
 1. A **team contract**
 2. A **project scope statement**
 3. A work breakdown structure (**WBS**)
 4. A **project schedule**, in the form of a Gantt chart with all dependencies and resources entered
 5. A **list of prioritized risks** (part of a risk register)



Planning Processes and Outputs (1)

Knowledge Area	Planning Process	Outputs
<i>Project Integration Management</i>	Develop project management plan	Project management plan
<i>Project Scope Management</i>	Plan scope management	Scope management plan Requirements management plan
	Collect requirements	Requirements documentation Requirements traceability matrix
	Define scope	Project scope statement Project documents updates
	Create WBS	Scope baseline Project documents updates
<i>Project Time Management</i>	Plan schedule management	Schedule management plan
	Define activities	Activity list Activity attributes Milestone list Project management plan updates
	Sequence activities	Project schedule network diagrams Project documents updates
	Estimate activity resources	Activity resource requirements Resource breakdown structure Project documents updates
	Estimate activity durations	Activity duration estimates Project documents updates
	Develop schedule	Schedule baseline Project schedule Schedule data Project calendars Project management plan updates Project documents updates



Planning Processes and Outputs (2)

<i>Project Cost Management</i>	Plan cost management	Cost management plan
	Estimate costs	Activity cost estimates Basis of estimates Project documents updates
	Determine budget	Cost baseline Project funding requirements Project documents updates



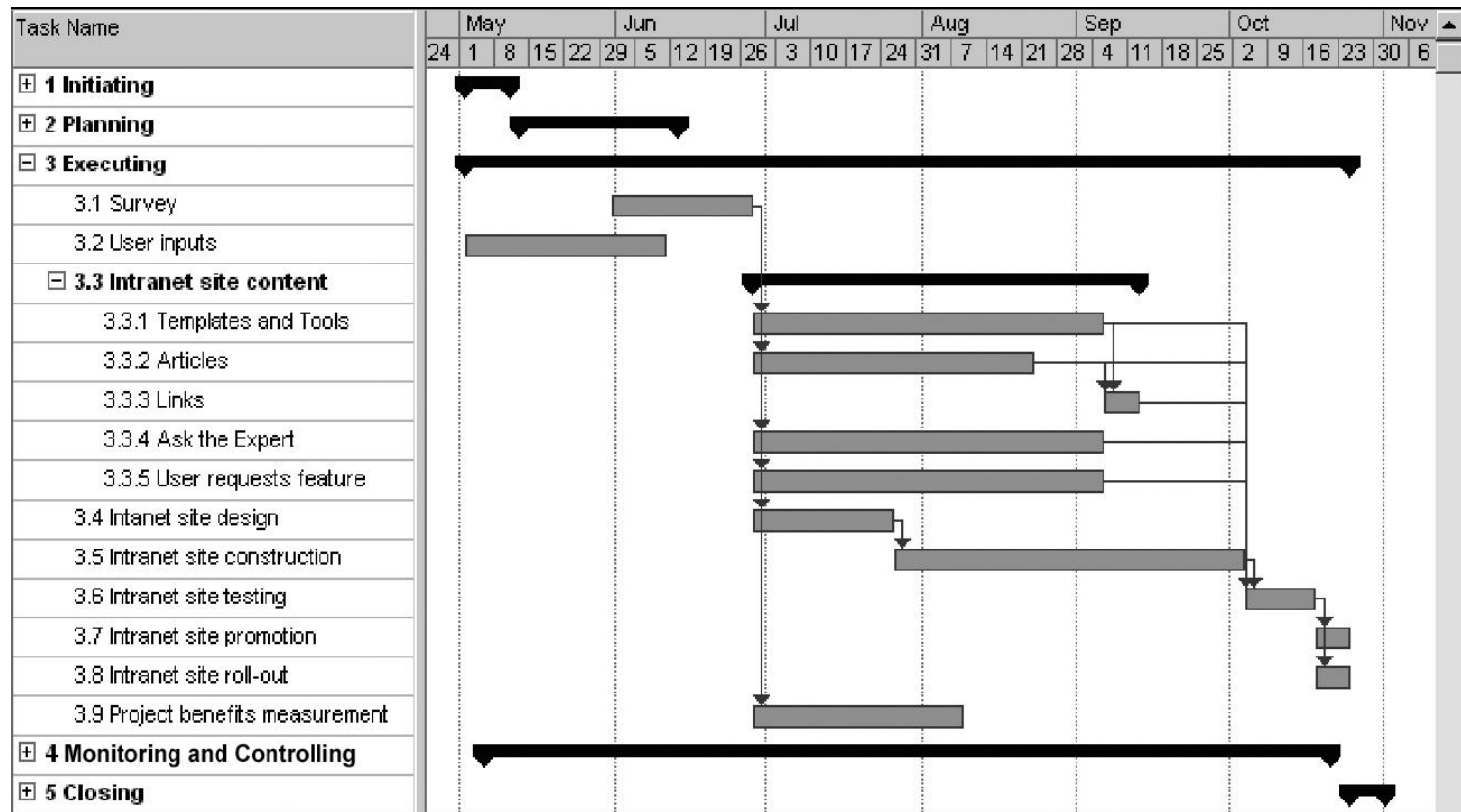


Planning Processes and Outputs (3)

Knowledge Area	Planning Process	Outputs
<i>Project Quality Management</i>	Plan quality management	Quality management plan Process improvement plan Quality metrics Quality checklists Project documents updates
<i>Project Human Resource Management</i>	Plan human resource management	Human resource plan
<i>Project Communications Management</i>	Plan communications management	Communications management plan Project documents updates
<i>Project Risk Management</i>	Plan risk management	Risk management plan
	Identify risks	Risk register
	Perform qualitative risk analysis	Project documents updates
	Perform quantitative risk analysis	Project documents updates
<i>Project Procurement Management</i>	Plan risk responses	Project management plan updates Project documents updates
	Plan procurement management	Procurement management plan Procurement statement of work Procurement documents Source selection criteria Make-or-buy decisions Change requests
<i>Project Stakeholder Management</i>	Plan stakeholder management	Stakeholder management plan Project documents updates



Gantt Chart





List of Prioritized Risks

RANKING	POTENTIAL RISK
1	Lack of inputs from internal consultants
2	Lack of inputs from client representatives
3	Security of new system
4	Outsourcing/purchasing for the article retrieval and “Ask the Expert” features
5	Outsourcing/purchasing for processing online payment transactions
6	Organizing the templates and examples in a useful fashion
7	Providing an efficient search feature
8	Getting good feedback from Michael Chen and other senior consultants
9	Effectively promoting the new system
10	Realizing the benefits of the new system within one year



Project Executing

- Usually **takes the most time and resources** to perform project execution
- Project managers must **use their leadership skills to handle the many challenges** that occur during project execution
- Table 3-11 on p. 108 **lists the executing processes and outputs**; many project **sponsors and customers focus on deliverables** related to providing the products, services, or results desired from the project
- A **milestone report** (example on pp. 109-110) can **help focus on completing major milestones**

Executing Processes and Outputs (1)

Knowledge Area	Executing Process	Outputs
<i>Project Integration Management</i>	Direct and manage project work	Deliverables Work performance data Change requests Project management plan updates Project documents updates
<i>Project Quality Management</i>	Perform quality assurance	Change requests Project management plan updates Project documents updates Organizational process assets updates
<i>Project Human Resource Management</i>	Acquire project team	Project staff assignments Resource calendars Project management plan updates
	Develop project team	Team performance assessments Enterprise environmental factor updates
	Manage project team	Change requests Project management plan updates Project documents updates Enterprise environmental factors updates Organizational process assets updates

Executing Processes and Outputs (2)

<i>Project Communications Management</i>	Manage communications	Project communications Project documents updates Project management plan updates Organizational process assets updates
<i>Project Procurement Management</i>	Conduct procurements	Selected sellers Agreements Resource calendars Change requests Project management plan updates Project documents updates
<i>Project Stakeholder Management</i>	Manage stakeholder engagement	Issue log Change requests Project management plan updates Project documents updates Organizational process assets updates

Part of Milestone Report

Milestone	Date	Status	Responsible	Issues/ Comments
Initiating				
Stakeholders identified	May 2	Completed	Erica and Joe	
Project charter signed	May 10	Completed	Erica	
Project kick-off meeting held	May 13	Completed	Erica	Went very well
Planning				
Team contract signed	May 13	Completed	Erica	
Scope statement completed	May 27	Completed	Erica	
WBS completed	May 31	Completed	Erica	
List of prioritized risks completed	June 3	Completed	Erica	Reviewed with sponsor and team
Schedule and cost baseline completed	June 13	Completed	Erica	
Executing				
Survey completed	June 28		Erica	Poor response so far!



Project Monitoring and Controlling

- Involves **measuring progress toward project objectives**, monitoring deviation from the plan, and taking correction actions
- Affects **all other process groups** and occurs during **all phases of the project life cycle**
- **Outputs** include **performance reports**, **change requests**, and **updates** to various plans

Monitoring & Controlling Processes and Outputs (1)

Knowledge Area	Monitoring and Controlling Process	Outputs
<i>Project Integration Management</i>	Monitor and control project work	Change requests Work performance reports Project management plan updates Project documents updates
	Perform integrated change control	Approved change requests Change log Project management plan updates Project documents updates
<i>Project Scope Management</i>	Validate scope	Accepted deliverables Change requests Work performance information Project documents updates
	Control scope	Work performance information Change requests Project management plan updates Project documents updates Organizational process assets updates

Monitoring & Controlling Processes and Outputs (2)

<i>Project Time Management</i>	Control schedule	Work performance information Schedule forecasts Change requests Project management plan updates Project documents updates Organizational process assets updates
<i>Project Cost Management</i>	Control cost	Work performance information Cost forecasts Change requests Project management plan updates Project documents updates Organizational process assets updates
<i>Project Quality Management</i>	Control quality	Quality control measurements Validated changes Validated deliverables Work performance information Change requests Project management plan updates Project documents updates Organizational process assets updates

Monitoring & Controlling Processes and Outputs (3)

Knowledge Area	Monitoring and Controlling Process	Outputs
<i>Project Communications Management</i>	Control communications	Work performance information Change requests Project documents updates Organizational process assets updates
<i>Project Risk Management</i>	Control risks	Work performance information Change requests Project management plan updates Project documents updates Organizational process assets updates
<i>Project Procurement Management</i>	Control procurements	Work performance information Change requests Project management plan updates Project documents updates Organizational process assets updates
<i>Project Stakeholder Management</i>	Control stakeholder engagement	Work performance information Change requests Project documents updates Organizational process assets updates



Project Closing

- Involves **gaining stakeholder and customer acceptance** of the final products and services
- Even **if projects are not completed**, they should be closed out to learn from the past
- **Outputs** include **project archives and lessons learned**, part of organizational process assets
- Most projects also include a **final report** and **presentation to the sponsor/senior management**



Closing Processes and Outputs

Knowledge Area	Closing Process	Outputs
<i>Project Integration Management</i>	Close project or phase	Final product, service, or result transition Organizational process assets updates
<i>Project Procurement Management</i>	Close procurements	Closed procurements Organizational process assets updates



Case Study: Project Management (Agile Approach)

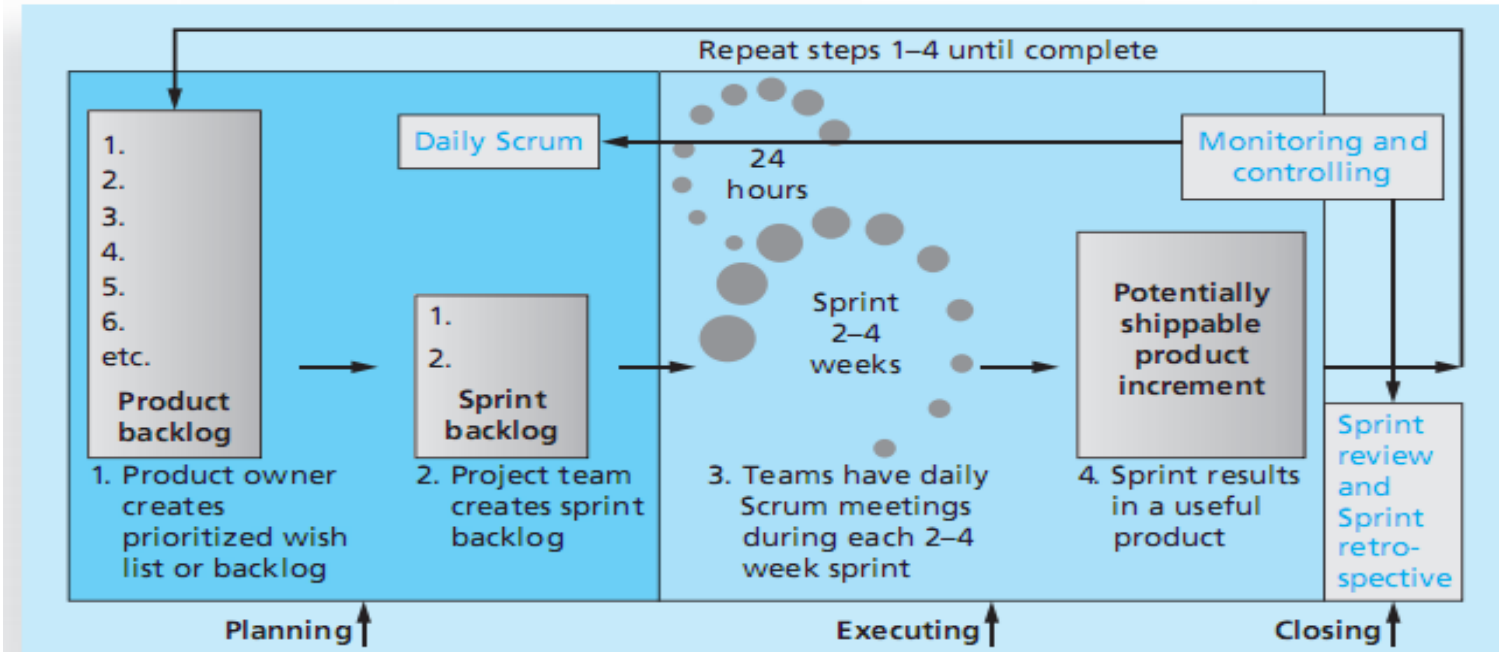
- Agile approach is often used for projects in which the business team cannot clearly express the scope early in the product life cycle, but the team does want to provide a potentially shippable product earlier rather than later
- When using agile techniques and its most popular method, Scrum, a team uses specific roles, artifacts, and ceremonies.



Scrum Roles, Artifacts, and Ceremonies

- **Scrum Roles**
 - Product Owner
 - Scrum Master
 - Scrum Team
- **Artifact**
 - Product backlogs
 - Sprint backlog
 - Burndown chart
- **Ceremony**
 - Sprint Planning Session
 - Daily Scrum
 - Sprint Review
 - Sprint retrospectives

Scrum Framework



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Scrum Framework and the Process Groups

Initiating:

- Determine roles
- Decide how many sprints will compose each release and the scope of software to deliver

Planning:

- Create product backlog
- Create sprint backlog
- Create release backlog
- Plan work each day in the daily Scrum
- Document stumbling blocks in a list

Executing:

- Complete tasks each day during sprints
- Produce a shippable product at the end of each sprint

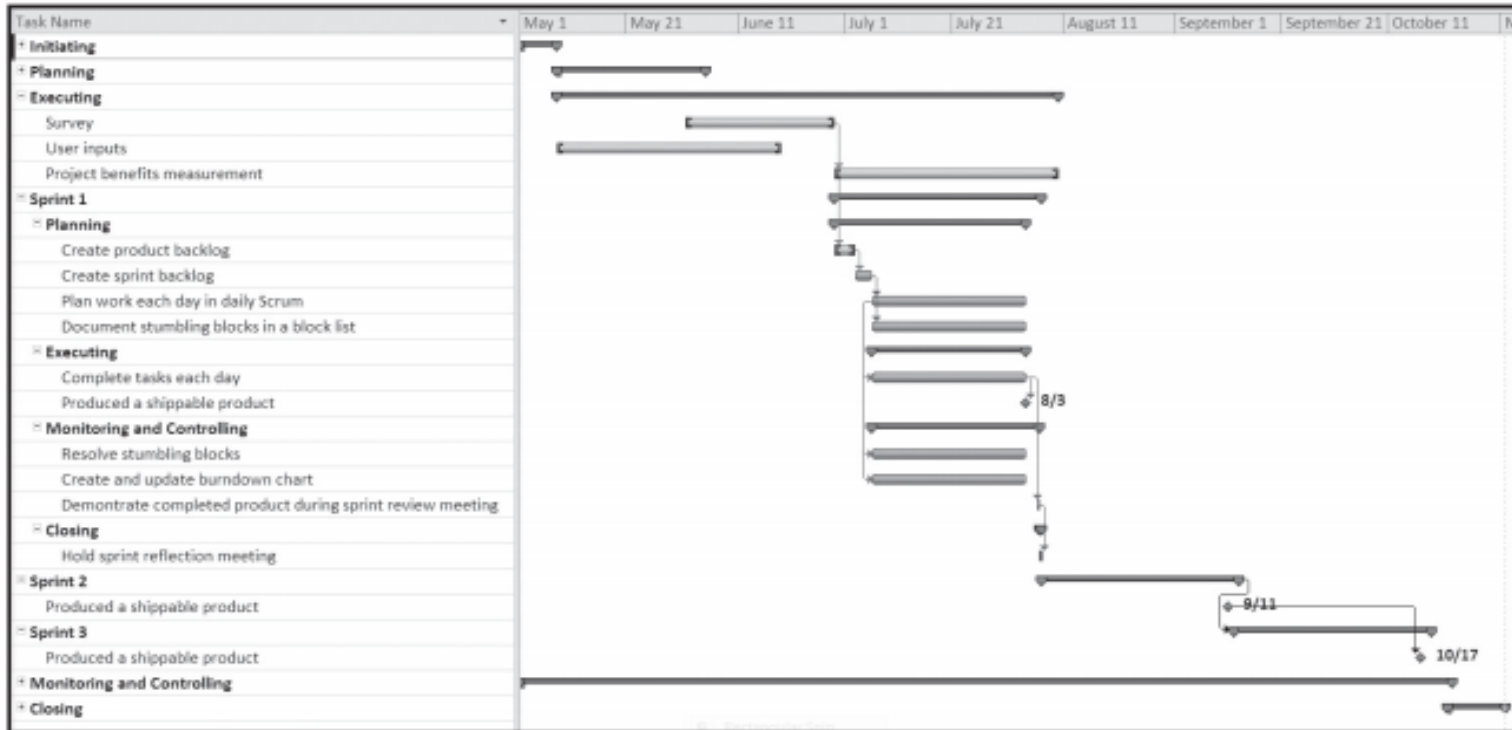
Monitoring and Controlling:

- Resolve issues and blockers
- Create and update burndown chart
- Demonstrate the completed product during the sprint review meeting

Closing:

- Reflect on how to improve the product and process during the sprint reflection meeting

Gantt chart using Scrum Approach (Planning)





Product and Sprint Backlogs (Planning)

Product Backlog

1. User story templates, samples, and point person
2. WBS templates, samples, and point person
3. Project schedule templates, samples, and point person
4. Ability to charge customers for some intranet products and services
5. Ability to collect user suggestions
6. Business case templates, samples, and point person
7. Ask the Expert feature
8. Stakeholder management strategy templates, samples, and point person
9. Risk register templates, samples, and point person
10. Etc.

Sprint Backlog

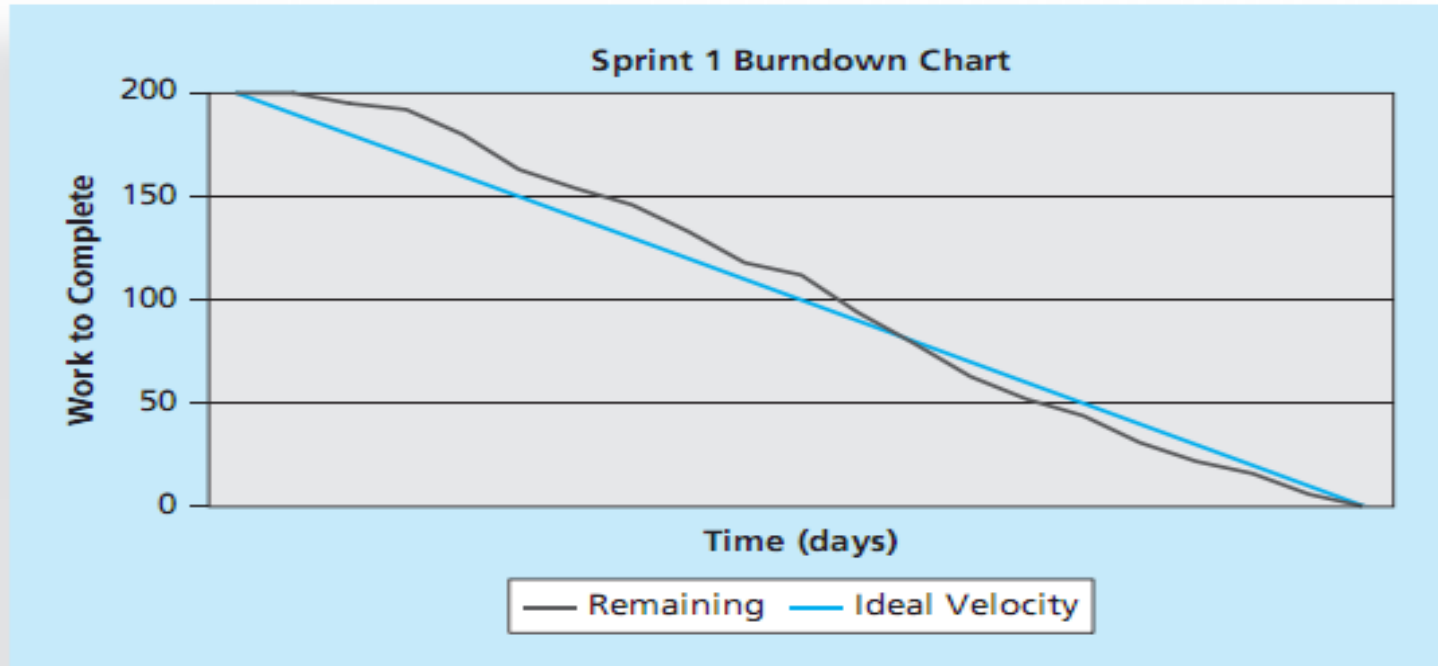
1. User story templates, samples, and point person
2. WBS templates, samples, and point person
3. Project schedule templates, samples, and point person
4. Ability to charge customers for some intranet products and services
5. Ability to collect user suggestions



Executing

- The team would complete tasks each day, as it would in the first version of this case,
- But by using an agile approach, the team would produce several iterations of a potentially shippable product.
- For example, at the end of the first sprint,

Burndown Chart (Monitoring)





Closing

- After the sprint review, the ScrumMaster leads a sprint retrospective
- The sprint retrospective is intended to answer two fundamental questions
 - What went well during the last sprint that we should continue doing?
 - What could we do differently to improve the product or process?

Templates by Process Group (1)

Template Name	Process Group	Chapter(s) Where Used	Application Software	File Name
Business Case	Pre-initiating	3	Word	business_case.doc
Business Case Financial Analysis	Pre-initiating	3, 4	Excel	business_case_financials.xls
Stakeholder Register	Initiating	3, 13	Word	stakeholder_register.doc
Stakeholder Management Strategy	Initiating	3, 13	Word	stakeholder_strategy.doc
Kick-off Meeting	Initiating	3	Word	kick-off_meeting.doc
Payback Chart	Initiating	4	Excel	payback.xls
Weighted Decision Matrix	Initiating	4, 12	Excel	wtd_decision_matrix.xls
Project Charter	Initiating	3, 4, 5	Word	charter.doc
Team Contract	Planning	3	Word	team_contract.doc
Requirements Traceability Matrix	Planning	5	Word	reqs_matrix.doc
Scope Statement	Planning	3, 4, 5	Word	scope_statement.doc
Statement of Work	Planning	12	Word	statement_of_work.doc
Request for Proposal	Planning	12	Word	rfp_outline.doc
Software Project Management Plan	Planning	4	Word	sw_project_mgt_plan.doc
Work Breakdown Structure	Planning	3, 5, 6	Word	wbs.doc

Templates by Process Group (2)

Gantt Chart	Planning, Executing	3, 5, 6	Project	Gantt_chart.mpp
Network Diagram	Planning, Executing	3, 6	Project	network_diagram.mpp
Project Cost Estimate	Planning	7	Excel	cost_estimate.xls
Earned Value Data and Chart	Monitoring and Controlling	7	Excel	earned_value.xls
Burndown Chart	Monitoring and Controlling	3, 7	Excel	burndownchart.xls
Quality Assurance Plan	Executing	8	Word	quality_assurance_plan.doc
Pareto Chart	Monitoring and Controlling	8	Excel	pareto_chart.xls
Project Organizational Chart	Planning, Executing	9	Power-Point	project_org_chart.ppt
Responsibility Assignment Matrix	Planning, Executing	9	Excel	ram.xls

Templates by Process Group (3)

Template Name	Process Group	Chapter(s) Where Used	Application Software	File Name
Resource Histogram	Planning, Executing	9	Excel	resource_histogram.xls
Communications Management Plan	Planning	10	Word	comm_plan.doc
Project Description (text)	Planning	10	Word	project_desc_text.doc
Project Description (Gantt chart)	Planning	10	Project	project_desc_Gantt.mpp
Milestone Report	Executing	3, 6	Word	milestone_report.doc
Change Request Form	Planning, Monitoring and Controlling	4	Word	change_request.doc
Progress Report	Monitoring and Controlling	3, 10	Word	progress_report.doc
Probability/Impact Matrix	Planning, Executing, Monitoring and Controlling	11	Power- Point	prob_impact_matrix.ppt
List of Prioritized Risks	Planning, Executing, Monitoring and Controlling	3, 11	Word	list_of_risks.doc
Risk Register	Planning, Monitoring and Controlling	11	Excel	risk_register.xls



Templates by Process Group (4)

Top 10 Risk Item Tracking	Planning, Monitoring and Controlling	11	Excel	top_10.xls
Breakeven/Sensitivity Analysis	Planning	11	Excel	breakeven.xls
Expectations Management Matrix	Monitoring and Controlling	13	Word	expectations.doc
Issue Log	Monitoring and Controlling	13	Word	issue_log.doc
Client Acceptance Form	Closing	10	Word	client_acceptance.doc
Lessons-Learned Report	Closing	3, 10	Word	lessons_learned_report.doc
Final Project Documentation	Closing	3, 10	Word	final_documentation.doc

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Templates

- lists the templates available on:
 - <http://www.kathyschwalbe.com>
 - <http://pmttexts.com/my-books/it-pm-book/>



Summary

- The **five project management process groups** are initiating, planning, executing, monitoring and controlling, and closing
- You can map the main activities of each process group to the **knowledge areas**
- Some organizations **develop their own** information technology project management methodologies
- The case study provides an **example of using the process groups**
- **Scrum** in Case Study **Align With Process Groups**



References

1. Kathy Schwalbe, **Managing Information Technology Projects 7th Edition**, *Course Technology, Cengage Learning*, 2012
2. A Guide to the Project Management Body of Knowledge: **PMBOK Guide 6th Edition**, *Project Management Institute*, 2017



Project Integration Management





Learning Objectives

- Describe an **overall framework for project integration management** as it relates to the other PM knowledge areas and the project life cycle
- Explain the **strategic planning process** and apply different **project selection methods**
- Explain the importance of **creating a project charter** to formally initiate projects
- Describe **project management plan** development, understand the content of these plans, and review approaches for creating them
- Explain **project execution**, its relationship to project planning, the factors related to successful results, and tools and techniques to assist in project execution



Learning Objectives (Cont)

- Describe the **process of monitoring and controlling project** work
- Understand the **integrated change control process**, planning for and managing changes on information technology projects, and developing and using a change control system
- Explain the **importance of developing and following good procedures** for closing projects
- Describe **how software can assist** in project integration management

OPENING CASE

Nick Carson recently became project manager of a critical biotech enterprise at his Silicon Valley company. This project involved creating the hardware and software for a DNA-sequencing instrument used in assembling and analyzing the human genome. The biotech project was the company's largest endeavor, and it had tremendous potential for future growth and revenues. Unfortunately, there were problems managing this large project. It had been underway for three years and had already gone through three different project managers. Nick had been the lead software developer on the project before top management made him the project manager. The CEO told him to do whatever it took to deliver the first version of the software for the DNA-sequencing instrument in four months and a production version in nine months. Negotiations for a potential corporate buyout with a larger company influenced top management's sense of urgency to complete the project.

Highly energetic and intelligent, Nick had the technical background to make the project a success. He delved into the technical problems and found some critical flaws that kept the DNA-sequencing instrument from working. Nevertheless, he was having difficulty in his new role as project manager. Although Nick and his team got the product out on time, top management was upset because Nick did not focus on managing all aspects of the project. He never provided them with accurate schedules or detailed plans of what was happening on the project. Instead of performing the work of project manager, Nick had taken on the role of software integrator and troubleshooter. Nick, however, did not understand top management's problem—he delivered the product, didn't he? Didn't they realize how valuable he was?



The Key to Project Success: Good Project Integration Management

- Project managers **must coordinate all of the other knowledge areas** throughout a **project's life cycle**
- Many new project managers have **trouble looking at the “big picture”** and want to focus on too many details (see opening case for a real example)
- Project integration management **is not the same thing as software integration**



Project Integration Management Processes

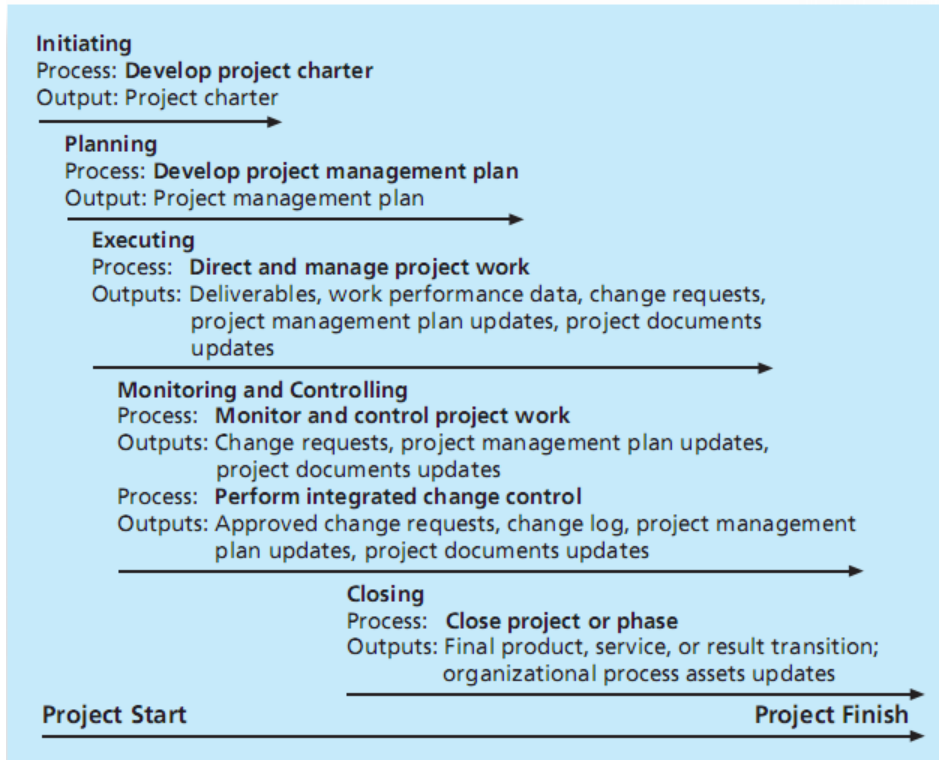
1. **Develop the project charter:** working with stakeholders to create the document that formally authorizes a project
2. **Develop the project management plan:** coordinating all planning efforts to create a consistent, coherent document
3. **Direct and manage project execution:** carrying out the project management plan by performing the activities included in it
4. **Monitor and control the project work:** overseeing project work to meet the performance objectives of the project
5. **Perform integrated change control:** coordinating changes that affect the project's deliverables and organizational process assets
6. **Close the project or phase:** finalizing all project activities to formally close the project or phase



Project Management Process Groups and Knowledge Area Mapping

Knowledge Area	Project Management Process Groups				
	Initiating	Planning	Executing	Monitoring and Controlling	Closing
<i>Project Integration Management</i>	Develop project charter	Develop project management plan	Direct and manage project execution	Monitor and control project work, Perform integrated change control	Close project or phase

Project Integration Management Summary

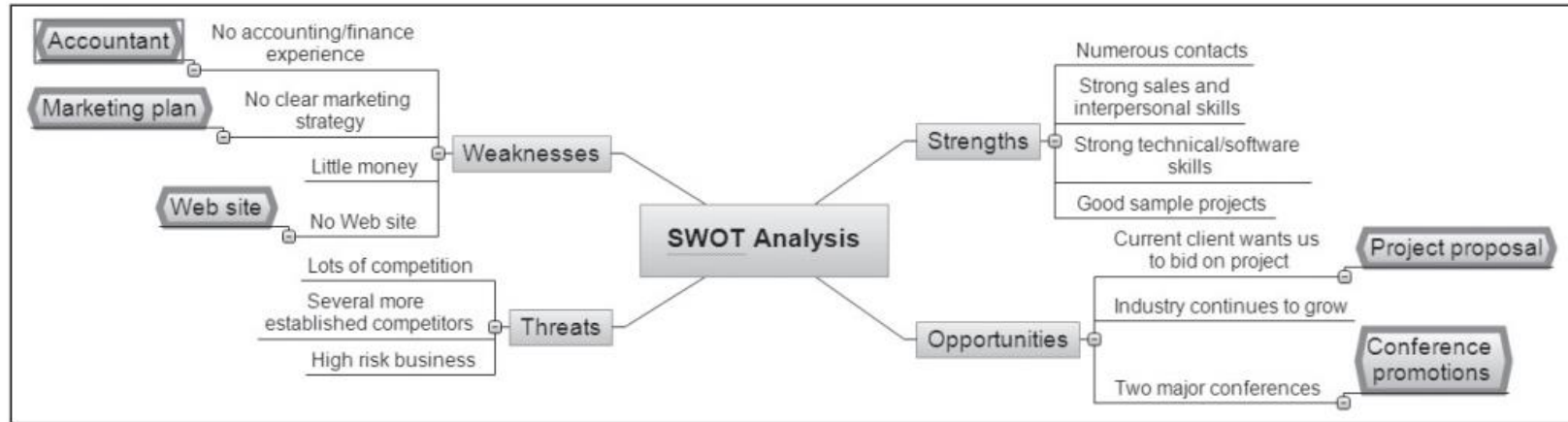




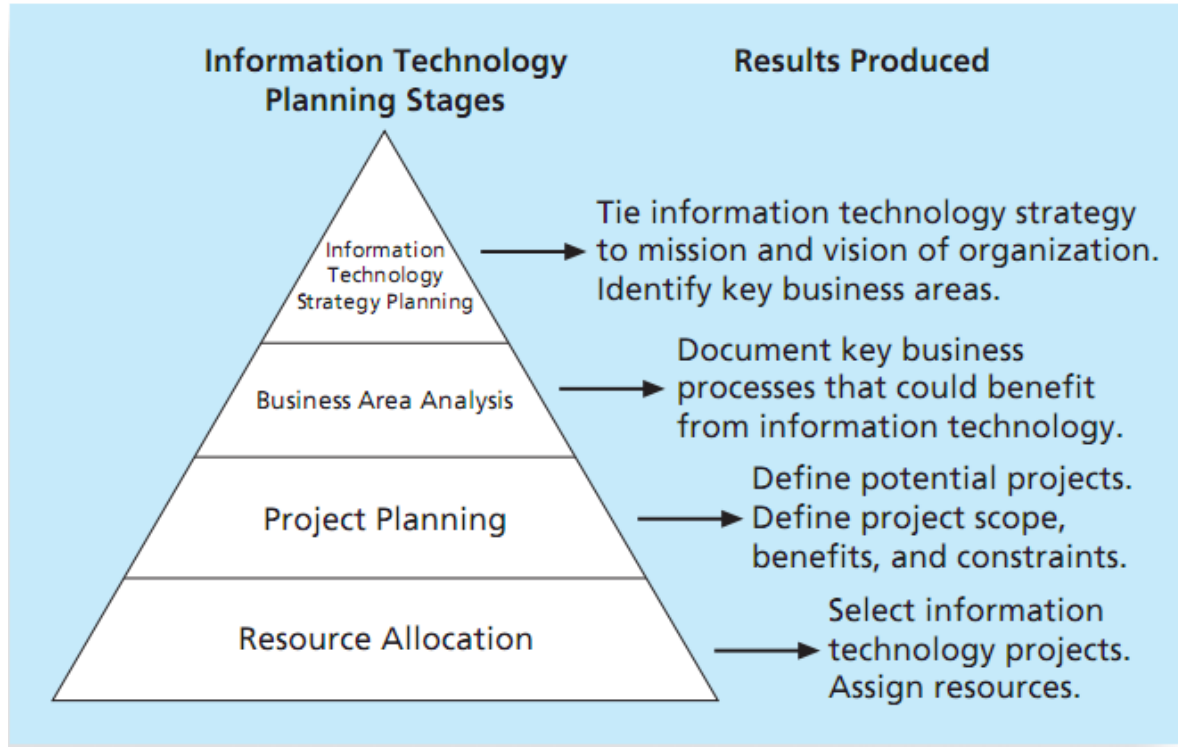
Strategic Planning and Project Selection

- **Strategic planning** involves determining **long-term objectives**, predicting **future trends**, and **projecting the need for new products** and services
- Organizations often perform a **SWOT analysis**
 - Analyzing **Strengths**, **Weaknesses**, **Opportunities**, and **Threats**
- As part of strategic planning, **organizations**:
 - **Identify potential projects**
 - Use realistic methods **to select which projects to work on**
 - Formalize project initiation by **issuing a project charter**

Mind Map of a SWOT Analysis to Help Identify Potential Projects



Information Technology Planning Process





Best Practice

- Only **one in seven product concepts comes to fruition**; why is it that some companies like Proctor & Gamble, Johnson and Johnson, Hewlett Packard, and Sony are **consistently successful in new product dev (NPD)**?
 - Because they use a **disciplined, systematic approach** to NPD projects based on best practices
- **Four important forces behind NPD success** include the following:
 1. A **product innovation** and technology strategy for the business
 2. Resource commitment and **focusing on the right projects**, or solid portfolio management
 3. An **effective, flexible, and streamlined** idea-to-launch process
 4. The right climate and **culture for innovation**, true **cross-functional teams**, and **senior management commitment** to NPD



Methods for Selecting Projects

- There are usually **more projects than available time and resources** to implement them
- **Methods for selecting projects** include:
 1. Focusing on broad **organizational needs**
 2. **Categorizing** information technology projects
 3. Performing net present value or other **financial analyses**
 4. Using a **weighted scoring model**
 5. Implementing a **balanced scorecard**



1. Focusing on Broad Organizational Needs

- It is often **difficult to provide strong justification for many IT projects**, but everyone agrees they have a high value
- “It is **better to measure gold roughly than to count pennies precisely**”
- Three important criteria for projects:
 - There is **a need** for the project
 - There are **funds available**
 - There’s a **strong will** to make the project succeed



2. Categorizing IT Projects

- One categorization is whether the **project addresses**:
 - A **problem**
 - An **opportunity**, or
 - A **directive**
- Another categorization is **how long it will take** to do and **when it is needed**
- Another is the overall priority of the project



3. Financial Analysis of Projects

- **Financial considerations** are often an important consideration in selecting projects
- Three primary methods for determining the projected financial value of projects:
 1. **Net present value** (NPV) analysis
 2. **Return on investment** (ROI)
 3. **Payback analysis**



4. Net Present Value Analysis

- Net present value (NPV) analysis is a **method of calculating the expected net monetary gain or loss from a project by discounting all expected future cash inflows and outflows to the present point in time**
- Projects with a **positive NPV** should be considered if financial value is a key criterion
- The higher the NPV, the better

Net Present Value Example

	A	B	C	D	E	F	G
1	Discount rate	10%					
2							
3	PROJECT 1	YEAR 1	YEAR 2	YEAR 3	YEAR 4	YEAR 5	TOTAL
4	Benefits	\$0	\$2,000	\$3,000	\$4,000	\$5,000	\$14,000
5	Costs	\$5,000	\$1,000	\$1,000	\$1,000	\$1,000	\$9,000
6	Cash flow	(\$5,000)	\$1,000	\$2,000	\$3,000	\$4,000	\$5,000
7	NPV →	\$2,316					
8		Formula =npv(b1,b6:f6)					
9							
10	PROJECT 2	YEAR 1	YEAR 2	YEAR 3	YEAR 4	YEAR 5	TOTAL
11	Benefits	\$1,000	\$2,000	\$4,000	\$4,000	\$4,000	\$15,000
12	Costs	\$2,000	\$2,000	\$2,000	\$2,000	\$2,000	\$10,000
13	Cash flow	(\$1,000)	\$0	\$2,000	\$2,000	\$2,000	\$5,000
14	NPV →	\$3,201					
15		Formula =npv(b1,b13:f13)					
16							
17							

Note that totals are equal, but NPVs are not because of the time value of money

JWD Consulting NPV Example

Discount rate	8%					
Assume the project is completed in Year 0			Year			
	0	1	2	3	Total	
Costs	140,000	40,000	40,000	40,000		
Discount factor	1	0.93	0.86	0.79		
Discounted costs	140,000	37,200	34,400	31,600	243,200	
Benefits	0	200,000	200,000	200,000		
Discount factor	1	0.93	0.86	0.79		
Discounted benefits	0	186,000	172,000	158,000	516,000	
Discounted benefits - costs	(140,000)	148,800	137,600	126,400	272,800	← NPV
Cumulative benefits - costs	(140,000)	8,800	146,400	272,800		
		▲				
ROI	→ 112%					
	Payback In Year 1					



NPV Calculations

- Determine **estimated costs and benefits** for the life of the project and the products it produces
- Determine the **discount rate** (check with your organization on what to use)
- Calculate the **NPV**
- Notes: Some organizations **consider the investment year as year 0, while others start in year 1**; some people enter costs as negative numbers, while others do not
 - Check with your organization for their preferences



Return on Investment

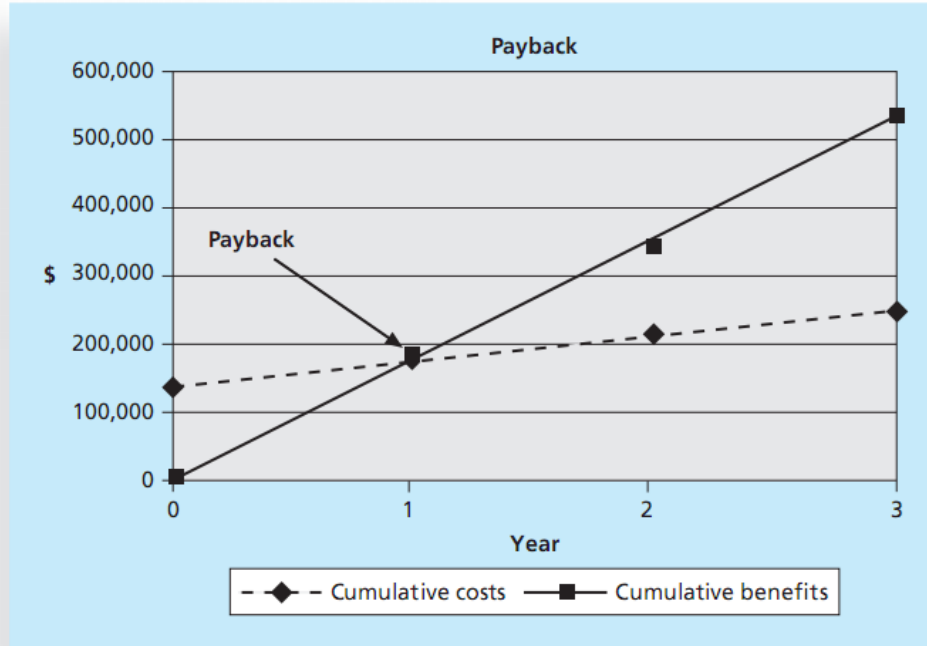
- Return on investment (ROI) is calculated by **subtracting the project costs from the benefits and then dividing by the costs**
 $ROI = (total\ discounted\ benefits - total\ discounted\ costs) / discounted\ costs$
- The higher the ROI, the better
- Many organizations have a **required rate of return or minimum acceptable rate of return on investment** for projects
- Internal rate of return (IRR) can be calculated by **finding the discount rate that makes the NPV equal to zero**



Payback Analysis

- Another important financial consideration is **payback analysis**
- The payback period is the **amount of time it will take to recoup, in the form of net cash inflows, the total dollars invested in a project**
- Payback occurs when **the net cumulative discounted benefits equals the costs**
- Many **organizations want IT projects to have a fairly short payback period**

Charting the Payback Period



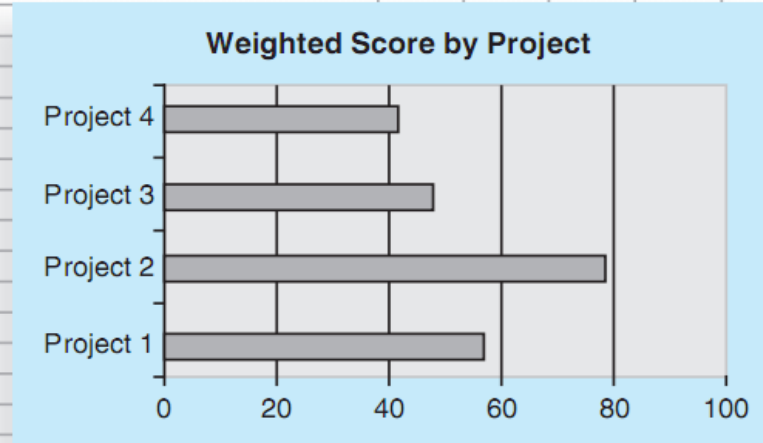


4. Weighted Scoring Model

- A weighted scoring model is a **tool that provides a systematic process for selecting projects based on many criteria**
 1. **Identify criteria important to the project selection process**
 2. **Assign weights (percentages) to each criterion** so they add up to 100%
 3. **Assign scores to each criterion** for each project
 4. Multiply the scores by the weights and **get the total weighted scores**
- The higher the weighted score, the better

Sample Weighted Scoring Model for Project Selection

	A	B	C	D	E	F
1	Criteria	Weight	Project 1	Project 2	Project 3	Project 4
2	Supports key business objectives	25%	90	90	50	20
3	Has strong internal sponsor	15%	70	90	50	20
4	Has strong customer support	15%	50	90	50	20
5	Uses realistic level of technology	10%	25	90	50	70
6	Can be implemented in one year or less	5%	20	20	50	90
7	Provides positive NPV	20%	50	70	50	50
8	Has low risk in meeting scope, time, and cost goals	10%	20	50	50	90
9	Weighted Project Scores	100%	56	78.5	50	41.5
10						
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5. Implementing a Balanced Scorecard

- Drs. Robert Kaplan and David Norton developed this approach **to help select and manage projects that align with business strategy**
- A balanced scorecard:
 - Is a **methodology that converts an organization's value drivers**, such as customer service, innovation, operational efficiency, and financial performance, to a series of defined metrics
- See www.balancedscorecard.org for more information

Balanced Scorecard Example

Mission: Provide responsive, professional finance and accounting services for the people who defend America

Vision:

Best Value to our customers

- World-class provider of finance and accounting services
- Trusted, innovative financial partner
- One Organization, One Identity
- Employer of choice, providing a progressive and professional work environment

Goals

- Fully satisfy customer requirements and aggressively resolve problems to deliver best value services
- Use performance metrics to drive best business practices and achieve high quality results
- Optimize the mix of our military, civilian, and contractor workforce
- Establish consultative relationships with leaders
- Deliver business intelligence to enable better decisions
- Ensure everyone is working towards the same vision and can connect what they're doing to make that vision a reality
- Embrace continuous learning for our workforce to ensure critical, high quality skill sets
- Develop the next generation of DFAS leadership

CUSTOMER PERSPECTIVE

- Improve client/customer satisfaction

FINANCIAL PERSPECTIVE

- Reduce cost to the client/customer
- Expand the use of competitive sourcing

INTERNAL PERSPECTIVE

- Improve and leverage quality
- Encourage innovation
- Deliver system solutions

GROWTH & LEARNING PERSPECTIVE

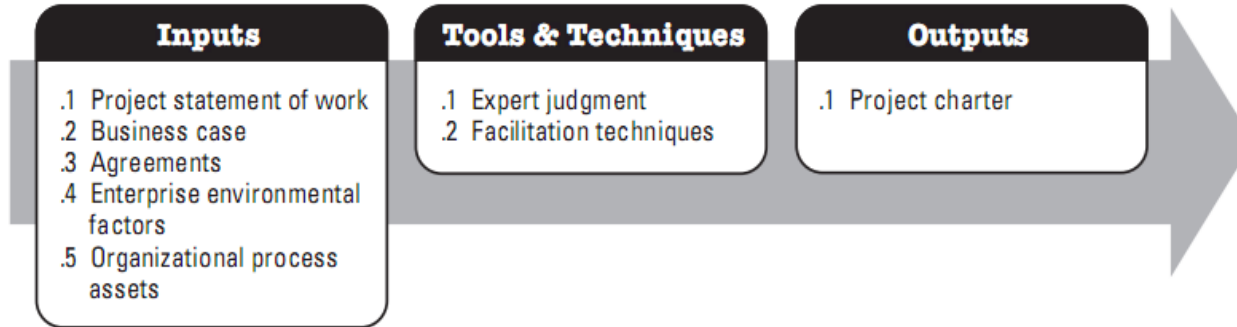
- Enhance employee competence
- Increase employee satisfaction
- Enhance ability to recruit and retain DFAS talent
- Develop climate for action



Project Charters

- After deciding what project to work on, it is important to let the rest of the organization know
- A **project charter** is a document that formally recognizes the existence of a project and provides direction on the project's objectives and management
- Key project stakeholders should sign a project charter to acknowledge agreement on the need and intent of the project
- a **signed charter is a key output of project integration management**

Develop Project Charter ITTO





Example: Project Charter

Project Title: DNA-Sequencing Instrument Completion Project

Date of Authorization: February 1

Project Start Date: February 1

Projected Finish Date: November 1

Key Schedule Milestones:

- Complete first version of the software by June 1
- Complete production version of the software by November 1

Budget Information: The firm has allocated \$1.5 million for this project, and more funds are available if needed. The majority of costs for this project will be internal labor. All hardware will be outsourced.

Project Manager: Nick Carson, (650) 949-0707, ncarson@dnaconsulting.com

Project Objectives: The DNA-sequencing instrument project has been underway for three years. It is a crucial project for our company. This is the first charter for the project, and the objective is to complete the first version of the software for the instrument in four months and a production version in nine months.

Main Project Success Criteria: The software must meet all written specifications, be thoroughly tested, and be completed on time. The CEO will formally approve the project with advice from other key stakeholders.



Example: Project Charter (cont)

Approach:

- Hire a technical replacement for Nick Carson and a part-time assistant as soon as possible.
- Within one month, develop a clear work breakdown structure, scope statement, and Gantt chart detailing the work required to complete the DNA sequencing instrument.
- Purchase all required hardware upgrades within two months.
- Hold weekly progress review meetings with the core project team and the sponsor.
- Conduct thorough software testing per the approved test plans.

ROLES AND RESPONSIBILITIES

Name	Role	Position	Contact Information
Ahmed Abrams	Sponsor	CEO	aaabrams@dnaconsulting.com
Nick Carson	Project Manager	Manager	ncarson@dnaconsulting.com
Susan Johnson	Team Member	DNA expert	sjohnson@dnaconsulting.com
Renyong Chi	Team Member	Testing expert	rchi@dnaconsulting.com
Erik Haus	Team Member	Programmer	ehaus@dnaconsulting.com
Bill Strom	Team Member	Programmer	bstrom@dnaconsulting.com
Maggie Elliot	Team Member	Programmer	melliot@dnaconsulting.com
Sign-off: (Signatures of all the above stakeholders)			
Ahmed Abrams		Nick Carson	
Susan Johnson		Renyong Chi	
Erik Haus		Bill Strom	
Maggie Elliot			
Comments: (Handwritten or typed comments from above stakeholders, if applicable)			
"I want to be heavily involved in this project. It is crucial to our company's success, and I expect everyone to help make it succeed." —Ahmed Abrams			
"The software test plans are complete and well documented. If anyone has questions, do not hesitate to contact me." —Renyong Chi			



Project Management Plans

- A **project management plan** is a document used to coordinate all project planning documents and help guide a project's execution and control
- Plans created in the other knowledge areas are subsidiary parts of the overall project management plan



Common Elements of a Project Management Plan

- Introduction or **overview of the project**
- Description of **how the project is organized**
- Management and technical processes used on the project
- **Work** to be done, **schedule**, and **budget** information



Sample Contents for a Software Project Management Plan (SPMP)

Major Section Headings	Section Topics
Overview	Purpose, scope, and objectives; assumptions and constraints; project deliverables; schedule and budget summary; evolution of the plan
Project Organization	External interfaces; internal structure; roles and responsibilities
Managerial Process Plan	Start-up plans (estimation, staffing, resource acquisition, and project staff training plans); work plan (work activities, schedule, resource, and budget allocation); control plan; risk management plan; closeout plan
Technical Process Plans	Process model; methods, tools, and techniques; infrastructure plan; product acceptance plan
Supporting Process Plans	Configuration management plan; verification and validation plan; documentation plan; quality assurance plan; reviews and audits; problem resolution plan; subcontractor management plan; process improvement plan

Source: IEEE Standard 1058–1998

Develop Project Management Plan ITTO



Figure 4-3. Develop Project Charter Data Flow Diagram



Project Execution

- Project execution involves managing and performing the work described in the project management plan
- The majority of time and money is usually spent on execution
- The application area of the project directly affects project execution because the products of the project are produced during execution



Coordinating Planning and Execution

- Project **planning and execution** are **intertwined and inseparable activities**
- Those **who will do the work** should help to plan the work
- Project managers **must solicit input from the team to develop realistic plans**



Providing Leadership and a Supportive Culture

- Project managers must **lead by example to demonstrate** the importance of creating and then following good project plans
- Organizational **culture can help project execution** by:
 - **Providing guidelines** and templates
 - **Tracking performance** based on plans
- Project managers may **still need to break the rules to meet project goals**, and senior managers must support those actions



Important Skills for Project Execution

- General management skills like leadership, communication, and political skills
- Product, business, and application area skills and knowledge
- Use of specialized tools and techniques



Project Execution Tools and Techniques

- **Expert judgment:** experts can help project managers and their teams make many decisions related to project execution
- **Project management information systems:** there are hundreds of project management software products available on the market today, and many organizations are moving toward powerful enterprise project management systems that are accessible via the Internet

Direct and Manage Project Work ITTO



Figure 4-6. Direct and Manage Project Work: Inputs, Tools and Techniques, and Outputs



Monitoring and Controlling Project Work

- **Changes are inevitable** on most projects, so it's important to develop and follow a process to monitor and control changes
- **Monitoring project** work includes **collecting**, **measuring**, and **disseminating** performance information
- A **baseline** is the **approved project management plan** plus approved changes



Integrated Change Control

- **Three main objectives** are:
 1. Influencing the factors that create changes to ensure that changes are beneficial
 2. Determining that a change has occurred
 3. Managing actual changes as they occur



Change Control on Information Technology Projects

- **Former view:** the project team should strive to do exactly what was planned on time and within budget
- **Problem:** stakeholders rarely agreed up-front on the project scope, and time and cost estimates were inaccurate
- **Modern view:** project management is a process of constant communication and negotiation
- **Solution:** changes are often beneficial, and the project team should plan for them



Change Control System

- A formal, documented process that describes when and how official project documents and work may be changed
- Describes who is authorized to make changes and how to make them



Change Control Board (CCB)

- A formal **group of people responsible for approving or rejecting changes** on a project
- CCBs provide **guidelines for preparing change requests, evaluate change requests, and manage the implementation** of approved changes
- Includes stakeholders from the entire organization



Making Timely Changes

- Some CCBs **only meet occasionally**, so it may **take too long for changes to occur**
- Some organizations **have policies in place for time-sensitive changes**
 - “**48-hour policy**” allows project team members to make decisions; then they have 48 hours to reverse the decision pending senior management approval
 - **Delegate changes to the lowest level possible**, but keep everyone informed of changes



Suggestions for Performing Integrated Change Control

View project management as a process of constant communication and negotiation

Plan for change

Establish a formal change control system, including a change control board (CCB)

Use effective configuration management

Define procedures for making timely decisions about smaller changes

Use written and oral performance reports to help identify and manage change

Use project management software and other software to help manage and communicate changes

Focus on leading the project team and meeting overall project goals and expectations





Configuration Management

- Ensures that the descriptions of the project's products are correct and complete
- Involves identifying and controlling the functional and physical design characteristics of products and their support documentation
- Configuration management specialists identify and document configuration requirements, control changes, record and report changes, and audit the products to verify conformance to requirements
- See www.icmhq.com for more information

Perform Integrated Change Control ITTO



Figure 4-10. Perform Integrated Change Control: Inputs, Tools & Techniques, and Outputs



Monitor and Control Project Work

- Monitor and Control Project Work is the **process of tracking, reviewing, and reporting the progress** to meet the performance objectives defined in the project management plan.
- The key benefit of this process is that it **allows stakeholders to understand the current state of the project**, the steps taken, and budget, schedule, and scope

Monitoring and Controlling ITTO

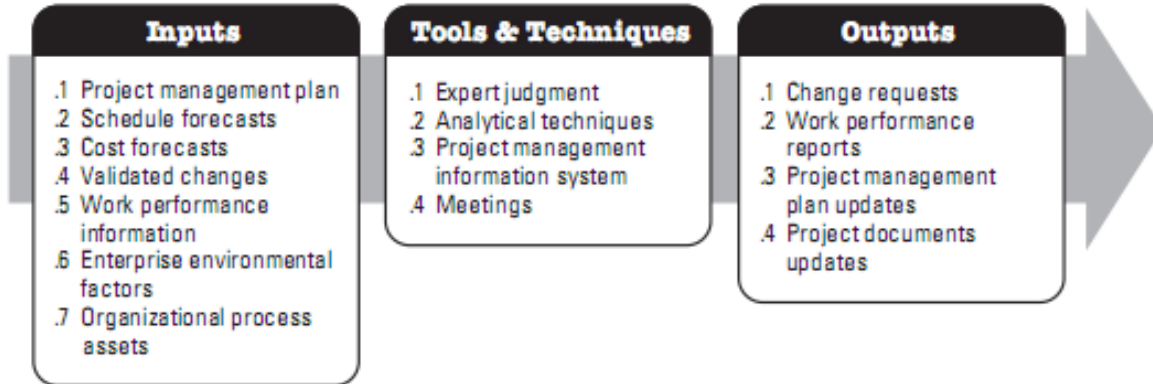


Figure 4-8. Monitor and Control Project Work: Inputs, Tools & Techniques, and Outputs



Closing Projects and Phases

- To close a project or phase, you must **finalize all activities** and **transfer the completed or cancelled work** to the appropriate people
- **Main outputs** include:
 - **Final product, service, or result** transition
 - Organizational process **asset updates**

Close Project or Phase ITTO



Figure 4-12. Close Project or Phase: Inputs, Tools & Techniques, and Outputs



Using Software to Assist in Project Integration Management

- **Several types of software** can be used to assist in project integration management
 - Documents can be created with **word processing software**
 - Presentations are created with **presentation software**
 - Tracking can be done with **spreadsheets or databases**
 - Communication software like **e-mail** and Web authoring tools facilitate communications
 - **Project management software can pull everything** together and show detailed and summarized information
 - Business Service Management (BSM) tools track the execution of business process flows



Summary

- Project integration management involves **coordinating all of the other knowledge areas throughout a project's life cycle**
- Main processes include:
 1. Develop **project charter**
 2. Develop **project management plan**
 3. Direct and manage **project execution**
 4. Monitor and control **project work**
 5. Perform **integrated change control**
 6. **Close the project** or phase



References

1. Kathy Schwalbe, **Managing Information Technology Projects 7th Edition**, *Course Technology, Cengage Learning*, 2012
2. A Guide to the Project Management Body of Knowledge: **PMBOK Guide 6th Edition**, *Project Management Institute*, 2017